*2026.04.14*

Introduction of Machine Learning

# PROPOSAL

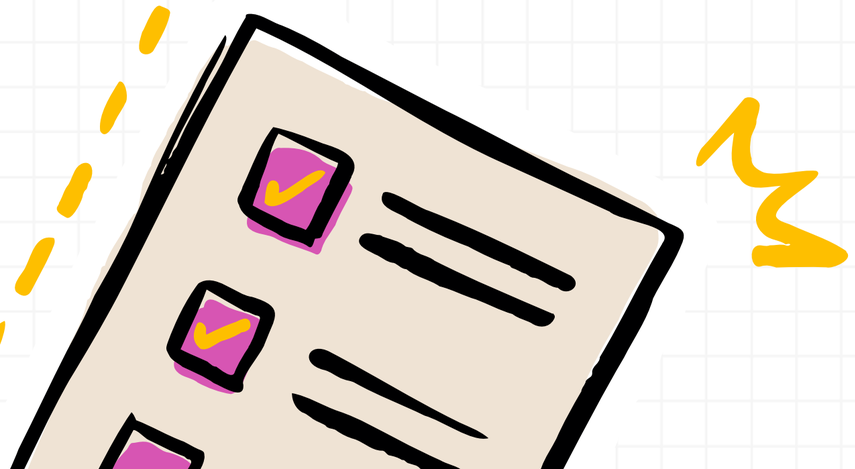
## *FOR FINAL PROJECT*

***Segment-Aware and Time-Aware Market Basket Analysis***

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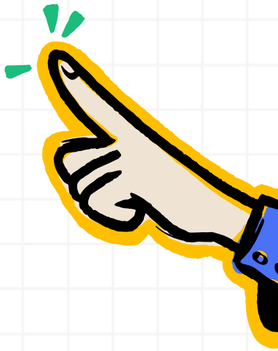


# TABLE OF *CONTENTS*



1. • **Objective, motivation:** what problem we want to solve
2. • **Dataset:** which dataset we want to use, how to collect it, what kinds of data cleaning will be required
3. • **Techniques, methods:** which ML method we want to implement, **what's the differences between my improvement and other existed code**
4. • **Expect experimental results:** what kinds of data we will demonstrate, what's the expected results

# 1 OBJECTIVE, MOTIVATION



MIS Background

Massive & Diverse E-Commerce Market

Online shopping platform

***We want to ...***

**use machine learning methods to solve inefficient marketing,**
**enabling precision marketing to target audiences and creating data into value.**

# 2 DATASET



- **Source:** Instacart Online Grocery Basket Analysis from Kaggle.
- **Complexity:** Over 3 million orders requiring the joining of 7+ relational tables.
- **Feature Construction:** Transforming raw transaction logs into user-level behavioral metrics, such as reorder rates.
- **Data Cleaning :** Handling missing values in order intervals and parsing product categories for analysis.

https://www.kaggle.com/datasets/yasserh/instacart-online-grocery-basket-analysis-dataset?select=orders.csv

TECHNIQUES, METHODS

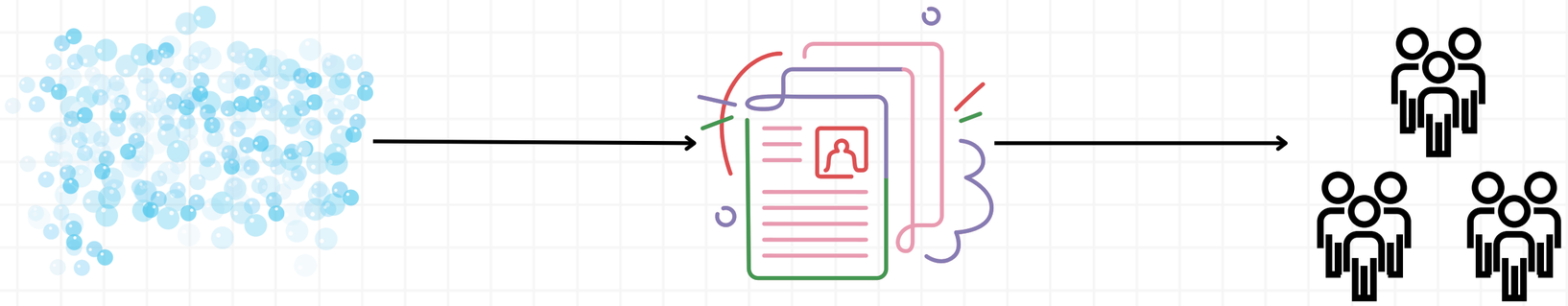
# 3 CUSTOMER SEGMENTATION

## *Feature Engineering（RFM + behavioral）*

Convert transaction data into user-level features (e.g., frequency, total items, reorder rate) to represent each customer for clustering.

## *K-means*

There are no predefined labels, we use K-means to cluster customers based on behavioral features. It is simple, efficient, and easy to interpret.





TECHNIQUES, METHODS

# MARKET BASKET ANALYSIS

## *Apriori / FP-growth*

Use association rule mining to discover frequently co-purchased products.

Analyze patterns within each customer segment.

## *Support, Confidence and Lift:*

**Support**

- Frequency of itemset in all transactions

$$Support(A, B) = \frac{\#\text{ transactions containing A and B}}{\text{total transactions}}$$

**Confidence**

- Probability of buying B given A

$$Confidence(A \rightarrow B) = \frac{Support(A, B)}{Support(A)}$$

**Lift**

- Strength of association between A and B

$$Lift(A \rightarrow B) = \frac{Confidence(A \rightarrow B)}{Support(B)}$$

3

TECHNIQUES, METHODS

# TIME-AWARE BASKET ANALYSIS

Key Feature!!

## *Apriori / FP-growth*

Apply association rule mining under different time conditions to discover time-dependent purchasing patterns.

## *Support, Confidence and Lift:*

Compute support, confidence, and lift within each time period to evaluate time-dependent associations.



# 4 EXPECT EXPERIMENTAL RESULTS

## Customer Segmentation

- Identify distinct customer groups
- Different purchasing behaviors across segments

## Basket Analysis

- Discover frequently co-purchased products
- Different patterns across customer segments

## Time-aware Analysis

- Purchasing behavior varies by time
- Different rules for weekday vs weekend
- Different rules for morning vs evening



# THANK *YOU*

Introduction of Machine Learning

Proposal For Final Project

Group I

Segment-Aware and Time-Aware Market Basket Analysis

